

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1741

Glen Oaks 138 kV
November 21, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1741. The Interconnection Customer proposes a 267.3 MW solar photovoltaic facility interconnecting at the Glen Oaks 138 kV substation in the Virginia area. The request was received on May 10, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 497 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 15 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$30,862,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: WESTON-GLEN OAKS SUBSTATION 138kV
Contingency: S TEXA-GLEN OAKS 138kV L
Loading: 155.4% (Rating: 266 MVA)
DFAX: 100.0%
Pre-project loading: 55.0%
2. Facility: WESTON-CANBY 138kV
Contingency: SOUTH TEXA-GLEN OAKS 138kV
Loading: 138.2% (Rating: 286 MVA)
DFAX: 100.0%
Pre-project loading: 44.9%
3. Facility: GLEN OAKS-S TEXA 138kV
Contingency: BRUSHTAVERN-RUSTBURG 138kV L
Loading: 132.4% (Rating: 394 MVA)
DFAX: 80.6%
Pre-project loading: 77.8%
4. Facility: CANBY-EVINGTON 138kV
Contingency: SOUTH TEXA-GLEN OAKS 138kV
Loading: 119.2% (Rating: 313 MVA)
DFAX: 100.0%
Pre-project loading: 33.8%
5. Facility: DESERT OUTPOST-S TEXA 138kV
Contingency: FRANKLIN-ELK GROVE-S EXA 138kV
Loading: 116.2% (Rating: 610 MVA)
DFAX: 51.2%
Pre-project loading: 93.8%

6. Facility: PINAS 9403-9405-EVINGTON 138kV Line
Contingency: S TEXA-GLEN OAKS 138kV
Loading: 114.2% (Rating: 321 MVA)
DFAX: 100.0%
Pre-project loading: 30.8%
7. Facility: FRANKLIN-ELK GROVE SUBSTATION-NEW LONDON SUBSTATION 138kV Line
Contingency: BEAR-CLOVER 345kV line
Loading: 113.1% (Rating: 490 MVA)
DFAX: 32.1%
Pre-project loading: 95.5%
8. Facility: FRANKLIN-ELK GROVE-BANNING 138kV
Contingency: FRANKLIN-ELK GROVE-SOUTH TEXA 138kV
Loading: 108.9% (Rating: 283 MVA)
DFAX: 24.2%
Pre-project loading: 86.0%
9. Facility: YANKEE DOODLE-ROCK ISLAND DAM 138kV
Contingency: REACH-BLUFF CITY 138kV line
Loading: 108.4% (Rating: 217 MVA)
DFAX: 7.1%
Pre-project loading: 99.6%
10. Facility: S TEXA-FRANKLIN-ELK GROVE SUBSTATION 138kV L
Contingency: DESERT OUTPOST-SOUTH TEXA 138kV
Loading: 108.3% (Rating: 491 MVA)
DFAX: 38.8%
Pre-project loading: 87.1%
11. Facility: PINAS 9403-9405-SOUTH COLTON 138kV line
Contingency: GLEN OAKS-S TEXA 138kV
Loading: 108.1% (Rating: 331 MVA)
DFAX: 100.0%
Pre-project loading: 27.2%
12. Facility: BANNING-DESERT OUTPOST 138kV
Contingency: FRANKLIN-ELK GROVE-S TEXA 138kV
Loading: 107.8% (Rating: 298 MVA)
DFAX: 24.2%
Pre-project loading: 86.1%
13. Facility: DESERT OUTPOST-MONEL 138kV L
Contingency: RUSTBURG SUBSTATION-BRUSH TAVEN SUBSTATION 138kV
Loading: 105.3% (Rating: 460 MVA)
DFAX: 19.4%
Pre-project loading: 94.1%
14. Facility: MONEL-LA JOIE 138kV
Contingency: MONEL-CADELARIA 138kV
Loading: 103.4% (Rating: 178 MVA)
DFAX: 9.8%
Pre-project loading: 88.6%
15. Facility: RED HILL-TIMKEN MERCY 138kV
Contingency: FLUVANNA P.S-RED HILL 345kV
Loading: 100.4% (Rating: 304 MVA)
DFAX: 6.3%
Pre-project loading: 94.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
GLEN OAKS-WESTON 138kV L	Rebuild line with higher-capacity conductor	\$3,162,000	62% (\$1,961,000)
WESTON-CANBY 138kV	Rebuild line with higher-capacity conductor	\$12,458,000	62% (\$7,726,000)
SOUTH TEXA SUBSTATION-GLEN OAKS 138kV	Rebuild line with higher-capacity conductor	\$3,003,000	46% (\$1,388,000)
CANBY-EVINGTON 138kV	Rebuild line with higher-capacity conductor	\$14,918,000	62% (\$9,252,000)
DESERT OUTPOST-S TEXA 138kV	Rebuild line with higher-capacity conductor	\$2,035,000	30% (\$616,000)
PINAS 9403-9405-EVINGTON 138kV line	Rebuild line with higher-capacity conductor	\$3,981,000	62% (\$2,469,000)
FRANKLIN-ELK GROVE SUBSTATION-NEW LONDON SUBSTATION 138kV Line	Rebuild line with higher-capacity conductor	\$4,122,000	18% (\$726,000)
FRANKLIN-ELK GROVE-BANNING 138kV	Reconductor line	\$783,000	22% (\$170,000)
YANKEE DOODLE-ROCK ISLAND DAM 138kV	Reconductor line	\$1,749,000	50% (\$867,000)
FRANKLIN-ELK GROVE-SOUTH TEXA 138kV line	Reconductor line	\$6,719,000	28% (\$1,896,000)
PINAS 9403-9405-SOUTH COLTON 138kV line	Reconductor line	\$1,728,000	62% (\$1,072,000)
BANNING-DESERT OUTPOT 138kV	Reconductor line	\$4,316,000	22% (\$939,000)
DESERT OUTPOST-MONEL 138kV L	Reconductor line	\$1,054,000	27% (\$285,000)
MONEL-LA JOIE 138kV	Reconductor line	\$5,503,000	26% (\$1,405,000)
RED HILL-TIMKEN MERCY 138kV	Reconductor line	\$956,000	9% (\$90,000)
	Total allocated to project		\$30,862,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.